A reservoir-constrained interaction model for graphite–CMC slurry rheology

Methods | Non-Brownian spheroids with hydrodynamic, frictional and adhesive interactions

1. Governing equations and mechanical interactions

The graphite-dominated suspension is represented by oblate spheroids of aspect ratio $a/b = 4$, particle volume fraction $\phi_p = 0.24$ and nominal volume-equivalent diameter $d_v = 8\,\mu$m. The small carbon-black fraction in the reference slurry is not resolved separately. The aqueous phase contains 90 kg mol^{-1} sodium carboxymethyl cellulose (NaCMC), matching the low-molecular-weight series of Gwag et al. [1]. At each total CMC concentration c_t, the model specifies three composition-dependent quantities: medium viscosity η_m, contact-excess adhesion W, and residual background amplitude α_{bg}. Particle geometry and the compressive contact law remain fixed.

In the non-Brownian, inertia-free approximation, translation and rotation obey

$$\mathbf{F}_i^H + \sum_{j\neq i}(\mathbf{F}_{ij}^C + \mathbf{F}_{ij}^{\mathrm{dir}} + \mathbf{F}_{ij}^{\mathrm{bg}}) = \mathbf{0}, \qquad \mathbf{T}_i^H + \sum_{j\neq i}(\mathbf{T}_{ij}^C + \mathbf{T}_{ij}^{\mathrm{dir}} + \mathbf{T}_{ij}^{\mathrm{bg}}) = \mathbf{0}. \tag{1}$$

Here H, C, dir and bg denote hydrodynamic resistance, mechanical contact, short-range contact attraction and noncontact background interaction. This decomposition follows the frictional-suspension framework [2, 3], with shape-dependent interaction geometry.

1.1. Hydrodynamic resistance

For imposed simple shear $\mathbf{u}^\infty = \dot\gamma z\,\mathbf{e}_x$, the generalized hydrodynamic load is

$$\begin{pmatrix}\mathbf{F}^H\\ \mathbf{T}^H\end{pmatrix} = -\eta_m(c_f)\widehat{\mathsf{R}}_{FU}(\mathcal{Q})\begin{pmatrix}\mathbf{U}-\mathbf{U}^\infty\\ \boldsymbol{\Omega}-\boldsymbol{\Omega}^\infty\end{pmatrix} + \eta_m(c_f)\widehat{\mathsf{R}}_{FE}(\mathcal{Q}) : \mathsf{E}^\infty. \tag{2}$$

The resistance contains isolated-spheroid drag and pair lubrication; $\mathcal{Q}$ denotes positions and orientations. Near-contact squeeze and tangential resistance scale respectively as η_m/h_{reg} and $\eta_m \ln(L_{ij}/h_{\mathrm{reg}})$, with geometry-dependent prefactors. The regularized gap is $h_{\mathrm{reg}} = \max(g_{ij}, 0) + \epsilon\bar r_{ij}$, where g_{ij} is signed surface separation, $r_i = d_{v,i}/2$, $\bar r_{ij} = (r_i + r_j)/2$, and $\epsilon = 10^{-3}$. Thus $d_v = 8\,\mu$m gives a nominal regularization of 4 nm. This regularization does not move the mechanical contact surface $g_{ij} = 0$.

1.2. Compressive, sliding and rolling contacts

For overlap $\delta_{ij} = \max(-g_{ij}, 0)$, the compressive load and friction bounds are

$$N_{ij} = k_n\delta_{ij}, \qquad \mathbf{F}_{ij}^t = -k_t\boldsymbol{\xi}_{ij}, \quad |\mathbf{F}_{ij}^t| \le \mu_s N_{ij}, \qquad \mathbf{M}_{ij}^r = -K_r\boldsymbol{\theta}_{ij}, \quad |\mathbf{M}_{ij}^r| \le \mu_r a_{ij}^* N_{ij}. \tag{3}$$

Here $a_{ij}^* = 2r_ir_j/(r_i + r_j)$; $\boldsymbol{\xi}_{ij}$ and $\boldsymbol{\theta}_{ij}$ are the elastic tangential and rolling displacements. Sliding and rolling constraints can strongly affect the rheology of nonspherical contact networks [4]. The friction limits use the compressive spring load, not an added adhesive-force magnitude. Attraction nevertheless increases frictional constraints by generating compressive preload through Eq. (1). The fixed contact parameters are calibrated against the nonadhesive suspension baseline, $\eta_r \simeq 200$, rather than interpreted as measured molecular friction coefficients.

The observable shear stress and viscosities are

$$\sigma_{xz} = \eta_m\dot\gamma + \sigma_{xz}^H + \sigma_{xz}^C + \sigma_{xz}^{\mathrm{dir}} + \sigma_{xz}^{\mathrm{bg}}, \qquad \eta_{\mathrm{app}} = \frac{\sigma_{xz}}{\dot\gamma}, \qquad \eta_r = \frac{\eta_{\mathrm{app}}}{\eta_m}. \tag{4}$$

Hydrodynamic stresslets and interaction-force virials supply the corresponding particle stresses. Consequently, changing CMC concentration changes both the viscous stress scale and the interaction-controlled structure.

2. Pure-graphite interaction potential

The attractive contribution for a particle pair is written as

$$U_{ij}^{\mathrm{int}} = U_{ij}^{\mathrm{dir}}(W, g_{ij}, \mathcal{Q}) + A_0 \alpha_{\mathrm{bg}}\, \mathcal{B}_{ij}(g_{ij}, \mathcal{Q}), \qquad W(0) = W_0, \quad \alpha_{\mathrm{bg}}(0) = 1. \tag{5}$$

The two amplitudes control distinct spatial ranges. In particular, W is the *contact excess* in addition to the background, not the total work of adhesion. Conservative forces and torques follow from spatial and orientational derivatives of the complete potential in Eq. (5).

2.1. Short-range contact excess

Let C_i and C_j be the local curvature tensors projected onto the common tangent plane. Their Derjaguin radius is $R_D = [\det(\mathsf{C}_i + \mathsf{C}_j)]^{-1/2}$. The contact-excess potential is

$$U_{ij}^{\mathrm{dir}} = \pi R_D W \begin{cases} 2g_{ij} - \ell, & g_{ij} < 0, \\ -\ell(1 - g_{ij}/\ell)^2, & 0 \le g_{ij} < \ell, \\ 0, & g_{ij} \ge \ell, \end{cases} \qquad \ell = 0.67 \text{ nm}. \tag{6}$$

At fixed local curvature, the tensile gap force has magnitude $2\pi R_D W$ in contact and decreases linearly to zero at $g_{ij} = \ell$. For general spheroid configurations, derivatives of R_D also contribute to the force and torque. The fixed range ℓ is a constitutive representation of unresolved near-contact adhesion; its value is not adjusted separately at each CMC concentration.

For the magnitude, we introduce a reference surface-energy scale,

$$W_0 = k_W(2\gamma_{GW}), \qquad \gamma_{GW} = 83 \text{ mJ m}^{-2}. \tag{7}$$

The reference follows graphene adhesion measurements in water [5]. The dimensionless k_W incorporates roughness, incomplete load-bearing contact and the mapping to the short-range kernel. It is determined by the pure-graphite sweep, not identified with a measured real-contact area fraction. The provisional inverse estimate $W_0 = 5.501$ mJ m^{-2} gives $k_W = 0.0331$.

2.2. Background dispersion interaction

The reference Hamaker amplitude is $A_0 = 9.9 \times 10^{-20}$ J, the long-wavelength graphite–water–graphite value reported by Li et al. [6]. The finite-spheroid kernel $\mathcal{B}_{ij}$ is held fixed while its effective amplitude is varied. In its near-contact limit, with $h = g_{ij} + h_0$ for $g_{ij} \ge 0$,

$$U_{ij}^{\mathrm{bg}} \simeq -\frac{A_0 \alpha_{\mathrm{bg}} R_D}{6h} + \frac{A_0 \alpha_{\mathrm{bg}} R_D \sigma_0^6}{1260 h^7}, \qquad h_0 = 2 \text{ nm}, \quad \sigma_0 = 0.4197 \text{ nm}. \tag{8}$$

For overlap, a linear continuation preserves the contact value and slope. At larger separation, the existing finite-spheroid continuation is retained. The small h^{-7} core term is part of that reference kernel; it is not an additional CMC repulsion. The planar attraction scale at the reference separation is

$$W_{\mathrm{bg},0} = \frac{A_0}{12\pi h_0^2}\left[1 - \frac{1}{30}\left(\frac{\sigma_0}{h_0}\right)^6\right] = 0.657 \text{ mJ m}^{-2}. \tag{9}$$

This is a planar free-energy density and a Derjaguin force scale, not a rule for adding the background potential depth to W.

Table 1: Fixed physical scales and calibration roles.

Quantity	Value / treatment	Basis and role
ϕ_p, a/b, d_v	0.24, 4, 8 μm	Fixed graphite geometry for the reference rheology.
η_w, M, DS	0.890 mPa s, 90 kg mol^{-1}, 0.7	Aqueous medium and CMC grade [1].
ℓ, h_0, A_0	0.67 nm, 2 nm, 9.9×10^{-20} J	Fixed near-contact scales; dispersion amplitude from Ref. [6].
$k_n, k_t, K_r, \mu_s, \mu_r$	Fixed across c_t	Mechanical baseline; stiffness chosen in the small-overlap regime.
W_0, W_4, $\alpha_{\mathrm{bg}}(c_t)$	Sweep-constrained	Effective interaction amplitudes; Table 2 gives starting estimates.

At fixed configuration and kinematics, the prescribed force and torque amplitudes scale as

$$\mathbf{F}^H \propto \eta_m(c_f), \qquad \mathbf{F}^{\mathrm{dir}} \propto W(c_t), \qquad \mathbf{F}^{\mathrm{bg}} \propto \alpha_{\mathrm{bg}}(c_t). \tag{10}$$

Figures 1 and 2 therefore specify the concentration dependence of the physical inputs; the mechanical contact coefficients remain unchanged.

3. CMC partition, medium viscosity and adsorption

All CMC concentrations are expressed per litre of liquid; 1 g L^{-1} = 1 kg m^{-3}. The mass balance and a minimal interpolation of the adsorption data are

$$c_t = c_a + c_f, \qquad c_a = \frac{q_\infty c_f}{K_d + c_f}, \qquad q_\infty = 2.8\ \mathrm{g\,L^{-1}}, \quad K_d = 0.04815\ \mathrm{g\,L^{-1}}. \tag{11}$$

These values reproduce $c_a = 2.7$ and $c_f = 1.3$ g L^{-1} at $c_t = 4$ g L^{-1} and the nominal adsorption plateau. Equation (11) interpolates the adopted Fig. 1(d) partition [1]; its intermediate values are not separate adsorption measurements. The surface excess Γ_m used below has units of mass per area and is distinct from c_a.

The continuous phase is treated as Newtonian over the shear-rate range considered. Its viscosity is assigned from the 90k-CMC solution data in Fig. 2(c) of Ref. [1]:

$$\eta_m(c_f) = \eta_w \left[1 + \mathcal{J}\{\eta_{\mathrm{sp}}^{\mathrm{exp}}\}(c_f)\right], \qquad \eta_{\mathrm{sp}}^{\mathrm{exp}} = \eta_{\mathrm{sol}}^{\mathrm{exp}}/\eta_w - 1. \tag{12}$$

Here $\mathcal{J}$ is a piecewise-linear interpolation after pooling small nonmonotonic digitization fluctuations. This uses the free, rather than total, CMC concentration. It is a measured-solution proxy for the slurry medium, without an extra fitted viscosity multiplier.

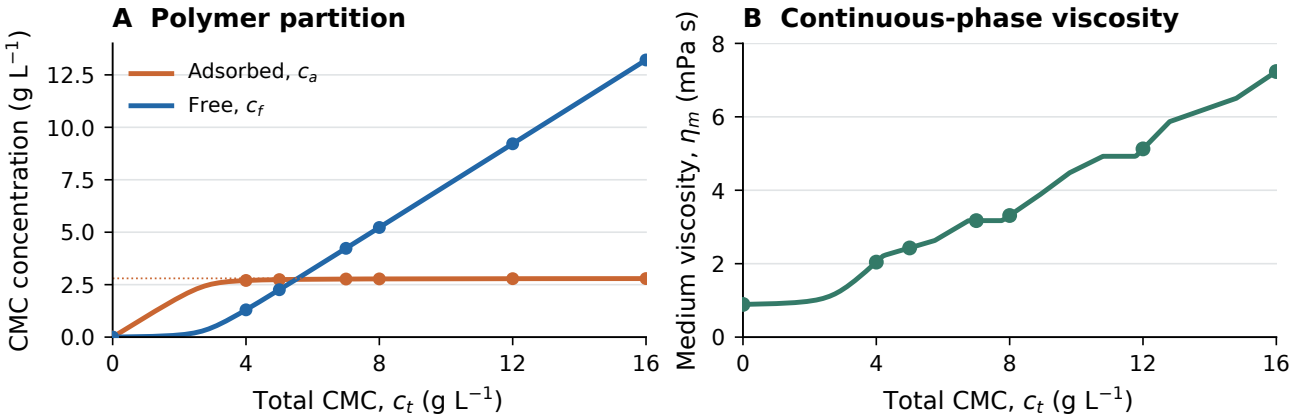


Figure 1: Composition-dependent medium. (A) Adopted adsorbed/free partition from Eq. (11). (B) Viscosity evaluated at the free-CMC concentration using Eq. (12). Symbols mark the concentration nodes in Table 2; lines show the interpolation.

At low dosage, adsorbed CMC suppresses direct carbon contact and changes graphite wettability, consistent with graphite–CMC surface measurements [7]. A minimal endpoint interpolation is

$$W_{\mathrm{ads}}(c_t) = W_0 - (W_0 - W_4)\min\left\{\frac{c_a(c_t)}{c_a(4)}, 1\right\}, \qquad W_4 = q_4 W_0. \tag{13}$$

The sweep determines q_4; its provisional value is 0.187. Equation (13) is a constitutive interpolation, not an identification of mass uptake with contact-area coverage. Above 4 g L^{-1}, the reservoir contribution in Section 4 determines the additional decrease.

Table 2: Concentration inputs and provisional inverse-rheology estimates at $\dot{\gamma} = 0.1\ \mathrm{s^{-1}}$.

c_t g L^{-1}	c_a g L^{-1}	c_f g L^{-1}	η_m mPa s	η_{exp} Pa s	σ_{exp} Pa	W mJ m^{-2}	α_{bg} —
0	0	0	0.890	7180	718	5.501	1
4	2.700	1.300	2.043	1850	185	1.028	1
5	2.742	2.258	2.428	1800	180	0.976	1
7	2.768	4.232	3.173	1600	160	0.771	1
8	2.774	5.226	3.312	1070	107	0.226	1
12	2.785	9.215	5.127	6.0	0.600	0	0.00586
16	2.790	13.210	7.234	2.4	0.240	0	0.00112

Experimental targets are digitized from the 90k series in Fig. 4(a), principally its inset [1]. W and α_{bg} are starting estimates transferred from the existing pure-graphite sweep, not completed CMC-specific 0.1 $\mathrm{s^{-1}}$ simulations. The transfer and the sequential fitting convention are defined in Section 5.

4. Free CMC as a chemical-potential reservoir

The physical hypothesis is that increasing free-CMC chemical potential raises the cost of expelling adsorbed polymer to form a direct graphite contact. Saturation of the adsorbed mass therefore need not imply saturation of contact free energy. The adsorption–force thermodynamics of Hall, Ash et al. and Pethica [8, 9, 10] provides the relevant sign independently of rheology.

Let $\omega(h, \mu_P)$ be the interaction grand potential per interfacial area, measured relative to two separated surfaces. For an electroneutral polymer component, including its associated counterions,

$$\frac{\partial \omega}{\partial \mu_P} = 2\Gamma_{P,\infty} - \Gamma_{P,h} \equiv b_P(h), \qquad \mathrm{d}\mu_P = \frac{M}{c_f}\mathrm{d}\Pi. \tag{14}$$

$\Gamma_{P,h}$ is the combined polymer excess in the interacting region and Π is the reservoir osmotic pressure along the specified composition path. For a direct contact that excludes adsorbed polymer, $b_P > 0$; its total reversible adhesion $\mathcal{W}_{\mathrm{dir}}$ decreases as

$$\frac{\mathrm{d}\mathcal{W}_{\mathrm{dir}}}{\mathrm{d}\ln c_f} = -b_m \frac{\mathrm{d}\Pi}{\mathrm{d}c_f}, \qquad b_m = M b_P. \tag{15}$$

The single-component osmotic form assumes a salt-free polymer/counterion dilution path with exchange-equilibrated interfaces. Added salt would require the corresponding multicomponent chemical potentials. For depletion from a nonadsorbing gap the excess difference can have the opposite sign; Eq. (15) specifically describes displacement of an adsorbed layer.

For counterion-dominated osmotic pressure, a useful first approximation is

$$\Pi \simeq \kappa c_f, \qquad \kappa = \frac{\varphi_{\mathrm{osm}} Z R T}{M}, \qquad Z = \frac{M\,\mathrm{DS}}{M_{\mathrm{repeat}}} \simeq 289. \tag{16}$$

Here $M_{\mathrm{repeat}} = 0.2182$ kg mol^{-1} for DS $= 0.7$. At $T = 298$ K, $\varphi_{\mathrm{osm}} = 0.3$–$0.5$ gives an osmotic scale of 3–52 kPa over $c_f = 1.3$–13.2 g L^{-1}. NaCMC counterion measurements support substantial charge renormalization [11]; φ_{osm} remains a system-specific osmotic input. When ions released from adsorbed CMC alter the reservoir composition, $\mathrm{d}\Pi/\mathrm{d}c_f$ must describe that actual medium.

While the background is fixed, the reduction in total direct-contact work is assigned to the contact-excess amplitude using the same rough-contact mapping as Eq. (7). With effective excluded mass $b_m^{\mathrm{eff}} \simeq 2k_W\Gamma_m$,

$$W(c_t) = \left[W_4 - K \ln \frac{c_f(c_t)}{c_{f,4}}\right]_+, \qquad K \simeq 2k_W\Gamma_m \frac{\varphi_{\mathrm{osm}} Z R T}{M}, \quad c_t \geq 4\ \mathrm{g\,L^{-1}}. \tag{17}$$

This is a rough-contact closure for the derivative, not an exact Young–Dupré identity for W. The conditional choice $\Gamma_m = 2$ mg m^{-2} and $\varphi_{\mathrm{osm}} = 0.4$ gives $K = 0.422$ mJ m^{-2} per e-fold; varying $\Gamma_m = 1$–3 and $\varphi_{\mathrm{osm}} = 0.3$–$0.5$ gives $K = 0.158$–0.791. This is a sensitivity scenario, not independent graphite bounds: CMC90k on iron nanoparticles gives 2.09 mg m^{-2} [12], whereas 330k CMC on natural graphite gives 0.24 mg m^{-2} [13]. The relevant Γ_m must therefore be constrained for the actual graphite surface.

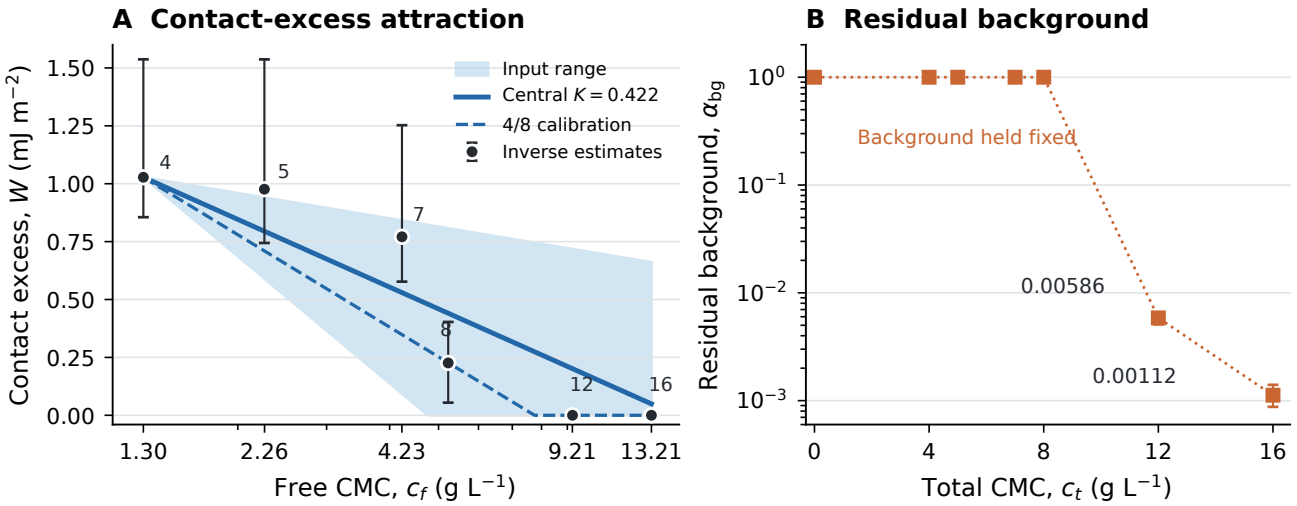


Figure 2: Attraction amplitudes. (A) Contact-excess estimates (symbols, labelled by total CMC) compared with Eq. (17). The band is the conditional Γ_m–φ_{osm} sensitivity range; the solid line uses its central values. The dashed line has $K = 0.576$ mJ m^{-2} calibrated at 4 and 8 g L^{-1}. (B) Provisional background amplitudes under the sequential inverse convention. Dotted connections are guides. Error bars denote digitization/window sensitivity, not statistical confidence intervals.

5. Coated contacts, residual attraction and parameter identification

The contact-excess law and the background coefficient have different physical meanings. Equation (17) represents the loss of favorable direct contact. Its clipping at zero removes a tensile *excess* term; it does not imply vanishing molecular dispersion or establish where the full direct and coated free energies cross. A coated state must instead be compared using its total interaction free energy,

$$w_{\mathrm{coat}}(h) = w_{\mathrm{vdW}}(h) + w_{\mathrm{el}}(h) + w_{\mathrm{polymer}}(h), \qquad W_{\mathrm{coat}} = \left[-\min_h w_{\mathrm{coat}}(h)\right]_+ . \tag{18}$$

Adsorbed CMC changes graphite wettability and interfacial structure [7]; an adsorbed layer coupled to free polymer can also generate a repulsive contribution [14]. These observations support a weakly adhesive coated branch. They do not require every polymer contribution to be repulsive. Layer compression is governed by the excess disjoining pressure, $\int(\Pi_{\mathrm{gap}} - \Pi_{\mathrm{bulk}})\,\mathrm{d}h$, so a larger bulk osmotic pressure alone is not a proof of stronger repulsion.

For a planar dispersion estimate,

$$w_{\mathrm{vdW}}(h) \simeq -\frac{A_0}{12\pi h^2}, \qquad \frac{|w_{\mathrm{vdW}}(h)|}{|w_{\mathrm{vdW}}(h_0)|} = \left(\frac{h_0}{h}\right)^2 . \tag{19}$$

Thus keeping two 4-nm layers between the carbon surfaces would increase h from 2 to 10 nm and reduce the nonretarded dispersion scale to 4%. Electrostatic/polymer contributions and retardation [15] can further reduce the net binding. The reference A_0 need not change. The layer thickness is an illustrative interfacial scale, not a concentration-dependent hard gap imposed on particle motion.

In the minimal rheological model, these unresolved coated interactions are represented by the effective amplitude α_{bg} in Eq. (5). Its role is to reproduce residual cohesion over the relevant contact configurations, rather than reproduce the full coated force curve at every gap. The estimate

$$\alpha_{\mathrm{bg}}\, W_{\mathrm{bg},0} \;\sim\; W_{\mathrm{coat}} \tag{20}$$

provides an interfacial scale check. Table 2 requires 3.85 and 0.737 μJ m^{-2} at 12 and 16 g L^{-1}, respectively. Distance alone would require equivalent nonretarded gaps of 26.1 and 59.7 nm; a few-nanometre coating therefore does not by itself account for the full decrease. The additional reduction is attributed to the net coated interaction and remains sweep-constrained. No separate electrostatic or steric force is added on top of an α_{bg} already representing that net attraction.

5.1. Sequential inverse calibration

At each concentration, η_m is fixed first from Eq. (12). The contact-excess amplitude is then decreased with $\alpha_{\mathrm{bg}} = 1$. Only when $W = 0$ remains too cohesive is α_{bg} reduced. This convention prioritizes contact weakening before changing the background; it is not a thermodynamic requirement that background suppression begin exactly at $W = 0$.

For the provisional estimates in Table 2, let $S(W, A_0)$ denote the existing pure-graphite stress sweep in water at $\dot{\gamma}_s = 100$ s^{-1}. The interaction-controlled stress is transferred approximately as

$$Y(W, A_0) \simeq S(W, A_0) - \beta\eta_w\dot{\gamma}_s, \qquad \sigma_{xz}(c_t, \dot{\gamma}) \simeq Y(W, A_0\alpha_{\mathrm{bg}}) + \beta\eta_m(c_f)\dot{\gamma}, \tag{21}$$

with $\beta = 200$. The contact branch uses interpolation of the measured sweep. Once $W = 0$, the starting estimate uses $Y(0, A_0\alpha_{\mathrm{bg}}) \simeq \alpha_{\mathrm{bg}} Y(0, A_0)$, where $Y(0, A_0) = 84.929$ Pa. The 0.1 s^{-1} CMC-specific sweeps replace these transfer approximations and determine the final amplitudes; no rate-independent yield law is assumed from a single transferred point.

5.2. Physical constraints on the final model

The pure-graphite sweep fixes k_W and the 4 g L^{-1} sweep fixes q_4. For $c_t \geq 4$, the constant-K law tests the chemical-potential mechanism with one additional slope, constrained by adsorption and osmotic scales. The central illustrative inputs give $W_8 = 0.441$, $W_{12} = 0.202$ and $W_{16} = 0.050$ mJ m^{-2}. The 4–8 g L^{-1} calibrated slope gives an excess-adhesion zero near 10.5 g L^{-1}; this zero is a calibrated consequence, not a parameter-free prediction. The 5 and 7 g L^{-1} targets therefore remain useful checks of the single-slope approximation.

At high CMC, small effective tensile amplitudes remove adhesive preload while retaining hydrodynamic dissipation and compressive sliding/rolling contacts. This allows a high-viscosity, weakly adhesive state without imposing a persistent repulsive barrier that prevents frictional contact. Holding the mechanical contact coefficients fixed across concentration is the minimal constitutive assumption. The contact-excess slope and residual background are calibrated separately, subject to the reservoir and coated-contact constraints above.

References

[1] E. Gwag, J. M. Won, J. W. Choi, S. Y. Kim, Interplay of adsorbed and free polymer governs microstructure and processability of battery slurries, *ACS Nano* **20** (2026) 25547–25561. 10.1021/acsnano.6c10201.

[2] R. Mari, R. Seto, J. F. Morris, M. M. Denn, Shear thickening, frictionless and frictional rheologies in non-Brownian suspensions, *J. Rheol.* **58** (2014) 1693–1724. 10.1122/1.4890747.

[3] A. Singh, S. Pednekar, J. Chun, M. M. Denn, J. F. Morris, From yielding to shear jamming in a cohesive frictional suspension, *Phys. Rev. Lett.* **122** (2019) 098004. 10.1103/PhysRevLett.122.098004.

[4] A. Singh, C. Ness, R. Seto, J. J. de Pablo, H. M. Jaeger, Shear thickening and jamming of dense suspensions: The "roll" of friction, *Phys. Rev. Lett.* **124** (2020) 248005. 10.1103/PhysRevLett.124.248005.

[5] C. D. van Engers, N. E. A. Cousens, V. Babenko, J. Britton, B. Zappone, N. Grobert, S. Perkin, Direct measurement of the surface energy of graphene, *Nano Lett.* **17** (2017) 3815–3821. 10.1021/acs.nanolett.7b01181.

[6] J.-L. Li, J. Chun, N. S. Wingreen, R. Car, I. A. Aksay, D. A. Saville, Use of dielectric functions in the theory of dispersion forces, *Phys. Rev. B* **71** (2005) 235412. 10.1103/PhysRevB.71.235412.

[7] J. Wu, I. Delcheva, Y. Ngothai, M. Krasowska, D. A. Beattie, Bubble–surface interactions with graphite in the presence of adsorbed carboxymethylcellulose, *Soft Matter* **11** (2015) 587–599. 10.1039/C4SM02380C.

[8] D. G. Hall, Thermodynamic treatment of some factors affecting the interaction between colloidal particles, *J. Chem. Soc., Faraday Trans. 2* **68** (1972) 2169. 10.1039/F29726802169.

[9] S. G. Ash, D. H. Everett, C. Radke, Thermodynamics of the effects of adsorption on interparticle forces, *J. Chem. Soc., Faraday Trans. 2* **69** (1973) 1256–1277. 10.1039/F29736901256.

[10] B. A. Pethica, Adsorption equilibria in surface force balance studies, *Colloids Surf. A* **105** (1995) 257–264. 10.1016/0927-7757(95)03324-6.

[11] E. Abbasi GharehTapeh, T. Watanabe, F. Horkay, C. Hou, C. G. Lopez, M. Hohenschutz, Counterion condensation, ion pairing and scattering properties of carboxymethyl cellulose with mono- and di-valent ions, *Cellulose* **33** (2026) 3833–3852. 10.1007/s10570-026-07021-3.

[12] H.-J. Kim, T. Phenrat, R. D. Tilton, G. V. Lowry, Fe^0 nanoparticles remain mobile in porous media after aging due to slow desorption of polymeric surface modifiers, *Environ. Sci. Technol.* **43** (2009) 3824–3830. 10.1021/es802978s.

[13] J.-H. Lee, U. Paik, V. A. Hackley, Y.-M. Choi, Effect of carboxymethyl cellulose on aqueous processing of natural graphite negative electrodes and their electrochemical performance for lithium batteries, *J. Electrochem. Soc.* **152** (2005) A1763–A1769. 10.1149/1.1979214.

[14] A. N. Semenov, A. A. Shvets, Theory of colloid depletion stabilization by unattached and adsorbed polymers, *Soft Matter* **11** (2015) 8863–8878. 10.1039/C5SM01365H.

[15] J. Gregory, Approximate expressions for retarded van der Waals interaction, *J. Colloid Interface Sci.* **83** (1981) 138–145. 10.1016/0021-9797(81)90018-7.